

A Post-hoc, Codec-agnostic Correction Layer for Conserving Global Invariants under Error-Bounded Lossy Compression

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Abstract

Error-bounded lossy compressors such as ZFP and SZ guarantee a pointwise error bound but do not preserve integrated quantities (total energy, mass, momentum), so downstream analyses that check a conservation law see a bias that grows with the compression ratio. This short note describes a simple post-processing layer that restores a chosen global invariant on top of any lossy backend, without modifying or recompiling it. For a positive quadratic invariant (energy) a single multiplicative factor, computed in closed form, matches the reconstruction’s value to that of the original; for linear or signed invariants, and for mass and energy jointly, a two-parameter affine correction does the same. This is not a new capability: reducing integral error under lossy compression has been addressed by error-bias correction in ZFP and by quantity-of-interest-preserving compression in the SZ line. The contribution here is narrow and practical: a trivial, backend-agnostic correction applied as post-processing, together with a measurement clarifying why it complements error-bias correction rather than competing with it. Statistical bias correction neutralizes the *mean* error and thus the linear part of the integral drift; a sum of squares retains a positive variance term that unbiased rounding does not remove, and the multiplicative correction removes it deterministically. We report the corrected measurements, state the method’s limitations plainly (it can exceed the codec’s pointwise bound by a bounded amount, and conservation is exact only to the precision of the stored factor), and note an in-loop use as future work. Code is available under Apache-2.0.

1 Introduction

The output volume of simulations and instruments has made lossy floating-point compression routine. Error-bounded compressors such as ZFP [1] and SZ [2] provide a guaranteed pointwise error bound, controlling local fidelity. They do not bound the error in *integrated* quantities computed from the field. Total kinetic energy, mass, and momentum are conservation checks whose drift under compression is a systematic bias rather than bounded noise, and that drift grows as compression becomes more aggressive.

This note documents a minimal post-processing response and, as importantly, its relationship to existing work. The construction is deliberately narrow: given any lossy backend, compute in closed form the correction that makes a chosen global invariant of the reconstruction match the original, store it, and apply it on decode. It is not a new codec, it does not change the compression ratio, and—as discussed below—it does not solve a problem the field has left open. Its value is as

a simple, backend-agnostic utility and as a clarification of where a deterministic correction helps beyond statistical error-bias reduction.

2 Related work

Reducing integral-quantity error under lossy compression is established. ZFP’s default truncation produces a biased error; rounding modifications (available in ZFP as the `ZFP_ROUND_FIRST` build option) neutralize that bias, which has been analyzed statistically [3] and applied to climate data [4]. Separately, quantity-of-interest-preserving compression in the SZ line [5] preserves derived quantities—including weighted sums (linear integrals) and polynomials (of which energy is a special case)—within user-specified bounds, by deriving per-point error bounds and integrating them into the compressor. Relative to these, the present layer is (i) applied as post-processing rather than inside the compressor, (ii) backend-agnostic, and (iii) deterministic and exact (to floating-point precision) for the specific invariant it targets, rather than statistical or bounded. It is correspondingly narrower: a single positive quadratic invariant for the multiplicative form, or mass and energy jointly for the affine form.

3 Method

Let $f \in \mathbb{R}^N$ be a field and $\hat{f} = C(f; \tau)$ its reconstruction under a lossy backend C at tolerance τ .

Multiplicative correction (positive quadratic invariant). For energy $E(x) = \sum_i x_i^2$, choose α so $E(\alpha\hat{f}) = E(f)$. Since $E(\alpha\hat{f}) = \alpha^2 E(\hat{f})$,

$$\alpha = \sqrt{E(f)/E(\hat{f})}. \quad (1)$$

The encoder stores α ; the decoder returns $\alpha\hat{f}$.

Affine correction (linear, signed, or joint invariants). The multiplicative form fails when the target invariant is signed or integrates to zero (the ratio is ill-defined). For such cases, and to conserve mass $M(x) = \sum_i x_i$ and energy $E(x)$ simultaneously, use $x' = a\hat{f} + b$ with $M(x') = M(f)$ and $E(x') = E(f)$. In closed form,

$$a = \sqrt{\frac{E(f) - M(f)^2/N}{E(\hat{f}) - M(\hat{f})^2/N}}, \quad b = \frac{M(f) - a M(\hat{f})}{N}, \quad (2)$$

i.e. a matches the centered energy (variance) and b matches the mean. This is well defined for signed or zero-valued M .

Why it complements bias correction. Writing $\hat{f}_i = f_i + e_i$, the energy drift is $E(\hat{f}) - E(f) = 2 \sum_i f_i e_i + \sum_i e_i^2$. If errors are symmetric and uncorrelated with f , the linear term $2 \sum f_i e_i$ vanishes in expectation, leaving the positive variance term $\sum e_i^2$. Error-bias correction neutralizes the mean error and hence the linear term (and thus most of the drift in a *linear* integral $\sum f_i$), but not $\sum e_i^2$. The multiplicative correction removes the residual deterministically.

Cost and guarantees. Storage is one scalar per invariant per field. The corrected pointwise error satisfies $\|f - \alpha \hat{f}\|_\infty \leq \|f - \hat{f}\|_\infty + |\alpha - 1| \|\hat{f}\|_\infty$, so the rescale can exceed the codec’s tolerance by up to $|\alpha - 1| \|\hat{f}\|_\infty$; it does not preserve the bound in general. Conservation is exact only to the precision of the stored factor.

4 Measurements

Tests use a synthetic 64^3 turbulence proxy (Taylor–Green flow plus multi-octave noise and localized bursts), `float32`, three seeds, with ZFP as the backend, built both with default rounding and with `ZFP_ROUND_FIRST` (bias correction). The field is a proxy, not a solution of the Navier–Stokes equations.

Energy, versus bias-corrected ZFP, matched on max error. Bias correction roughly halves the energy drift and improves the ratio, but leaves drift in the range 10^{-5} to 10^{-3} ; the multiplicative correction removes it to $\sim 10^{-10}$ (Table 1).

target max error	bias-corrected ZFP energy drift	+ correction
0.01	1.0×10^{-5}	9.6×10^{-11}
0.10	8.4×10^{-4}	2.9×10^{-9}

Table 1: Relative energy drift, bias-corrected ZFP with and without the multiplicative correction, matched on maximum absolute error (mean over 3 seeds).

Linear sum ($\sum f_i$). Consistent with the analysis, `ZFP_ROUND_FIRST` reduces the linear-sum drift by roughly $1.3\times$ to $4\times$ at matched max error—more than it reduces the energy drift—and the affine/additive correction drives it to machine precision ($\sim 10^{-16}$).

Tolerance. Across all tested tolerances the corrected reconstruction never exceeded ZFP’s requested tolerance, because ZFP’s achieved error sits at ~ 10 – 20% of the requested bound, leaving more headroom than the rescale ($|\alpha - 1| \leq 5.5 \times 10^{-3}$) consumes. This is a property of ZFP’s loose fixed-accuracy bound; a compressor with a tight bound (e.g. SZ) could be pushed over, and a clamped- α variant restores a hard guarantee at a small cost to conservation.

Precision floor. Conservation reaches $\sim 10^{-8}$ with a 4-byte (`float32`) stored factor and $\sim 5 \times 10^{-10}$ with an 8-byte factor. When the required correction falls below the stored factor’s unit in the last place (the near-lossless regime), the factor rounds to 1 and the correction is a no-op, which is benign since the uncorrected drift there is already at that floor.

5 Applications and limitations

The layer suits workflows that need a specific integral pinned after lossy compression without modifying the compressor—for archival where a conservation check must pass on the decompressed field, and, as a direction not evaluated here, for in-loop use inside memory-bandwidth-bound solvers (e.g. lattice-Boltzmann), where uncorrected drift in a conserved quantity can accumulate across timesteps.

Limitations: the multiplicative form targets one positive quadratic invariant; the general QoI problem is addressed more broadly by prior work [5]; results are on a synthetic proxy with one backend; the pointwise bound is not formally preserved (bounded overshoot); and “exact” means to the precision of the stored factor.

6 Conclusion

A trivial post-processing correction conserves a chosen global invariant on top of an unmodified lossy compressor—energy multiplicatively, mass and energy jointly via an affine form—at negligible cost. Its distinction from error-bias correction is specific: it removes the variance term in a sum of squares that unbiased rounding leaves behind, deterministically and per field. It is offered as a utility and a clarification, not a new method. Next steps are evaluation on measured data, an in-loop test in a conservation-sensitive solver, and an SZ backend where the tolerance interaction differs.

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Code and data availability

Reference code and benchmarks: <https://github.com/alikamp/conserved-field-compression> (Apache-2.0); the reported numbers reproduce from the included notebook.

References

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